

```
mydata = record.text
mysoup = BeautifulSoup(mydata)
for link in mysoup.find_all('a'):
    print(link.get('href'))
```

Using Python in cloud infrastructure

There are a number of service providers offering private or public cloud services. Listed below are a few prominent cloud and Big Data based service providers that provide options to code and deploy one's own applications:

- Amazon EC2
- IBM Bluemix
- Microsoft Azure
- Google
- Qubole

Cloud service providers make use of specialised packages and tools for coding in different languages. Python can be used on these cloud computing infrastructures.

On IBM Bluemix, the cloud services can be accessed using <https://www.bluemix.net>, and can be used once the credentials are verified. Cloud infrastructure can be used with the Cloud Foundry CLI (Command Line Interface).

After installation of Cloud Foundry (CF), one can communicate with the remote server. To check the version, the following command is used:

```
$ cf -v
```

The following instruction is used for pushing the application on the cloud server:

```
$ cf push MyApp -m 128M -b https://github.com/cloudfoundry/cf-buildpack-python.git
```

Programming Python for NoSQL databases

NoSQL databases are being used in social media applications and portals that process Big Data—in which huge, heterogeneous and unstructured data formats are handled. NoSQL databases are used for faster access of records from the big data set at the back-end. The *Aadhaar* card implementation in India is being done using NoSQL databases as huge amounts of information are involved, including text data, images, thumb impressions and iris detection. No traditional database system can handle data sets of different types (text, video, images, audio, thumb impressions, iris samples, etc) simultaneously.

Currently, a number of NoSQL databases are used for different types of portals, and these specialise in handling heterogeneous and unstructured data.

In traditional Web based implementations, RDBMS packages are deployed for database applications including Apache Derby, MySQL, Oracle, IBM DB2, Microsoft SQL Server, IBM Notes, PostgreSQL, SQLite, Sybase, etc. These are known as traditional SQL databases that comply with the ACID (atomicity, consistency, isolation and durability) properties. NewSQL is the

new-generation database engine that provides the scalability and performance of NoSQL systems for online transaction processing (OLTP) read-write workloads, while maintaining the ACID guarantees of a classical database system.

Nowadays, Web applications use unstructured data in heterogeneous formats including audio, video, text, streaming, signals, images, pixels and many others. In each file pattern, there are a number of file formats. For instance, in video, there are a number of file formats including MPEG, MP4, AVI, 3GP, WMV, OGG, FLV and others. In the same manner, image or graphics file formats include GIF, PNG, JPEG, PCX, BMP, TIFF and lots of others.

The major issue here is the compatibility of Web applications with all these file formats in different domains. Here's where the concept of NoSQL databases comes into play since they enable any type of file format to be processed and integrated in the Web applications. A NoSQL (Not Only SQL) database provides the system for storing and retrieving data, and is not modelled on the tabular relations methodology used in relational databases. The data structure in NoSQL databases is entirely different from that in traditional RDBMSs. Currently, the former are being rapidly adopted in Big Data and real-time Web applications.

There have been various approaches to classifying NoSQL databases, each with different categories and sub-categories. Because of the variety of approaches and overlaps it is difficult to get an overview of non-relational databases. A few categories are listed below:

Document	Apache CouchDB, Couchbase, MarkLogic, MongoDB
Column	Cassandra, Accumulo, Druid, HBase
Graph	InfiniteGraph, Allegro, Neo4J, OrientDB, Virtuoso, Stardog
Key-Value	MemcacheDB, Redis, Dynamo, Riak, FoundationDB, Aerospike, FairCom, c-treeACE

Programming Python - CouchDB

Apache CouchDB is one of the most popular open source databases, widely used as a document-oriented NoSQL database. Like most other NoSQL databases, CouchDB uses the JSON (JavaScript Object Notation) format to store data. The JSON format is the open standard data file format that is used as an alternate to XML to transmit data between multiple incompatible and heterogeneous servers.

The following example shows a sample JSON representation:

```
{
  "FirstName": "X",
  "LastName": "X",
  "LiveStatus": X,
  "Gender": X,
```